A REFINED ALGEBRAIZATION STEP: GLUING COHERENT SYSTEMS WITH PROPER SUPPORT

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 9849-9938, immediately following the statement at 9827-9847.

Editorial note. The statement itself references the situation of the section by \ref, and the proof cites several cross-chapter results; all are navigation pointers recorded in assembly.external_references.

Editorial note. The final properness argument is displayed in full inside the span (kernel support, G_1 support, H support, union lemma).

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 9897, 9906 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- The situation of Grothendieck's existence theorem. chapter situation situation-existence, cited in the statement and by equation-completion-functor-proper-over-A. Fixes the formal setup: the ideal I, the thickenings X_e, and the category Coh with support proper over A.
- Easy algebraization step. chapter lemma lemma-existence-easy, cited inside the proof. Provides the simpler algebraization statement that the refined argument reduces to.
- Inverse systems with kernel and cokernel annihilated by I. Coherent, remark coherent-remark-inverse-systems-kernel-cokernel-annihilated-by, cited inside the proof. Controls how a map of inverse systems behaves on kernels and cokernels under the thickenings.
- Union of closed subsets proper over the base. Derived Categories of Spaces, lemma spaces-perfect-lemma-union-closed-proper-over-base, cited at proof lines 9936-9937. Concludes properness of the support of the algebraized module from properness of the supports of its kernel and of H.

Source reference. Original reference 0888: focused-seeds/geometry/S9-spaces-existence-tricky/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original equation 08B8: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9710–9710. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 08BB: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9801–9801. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08B7: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9682–9682. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08AM: focused-seeds/geometry/S9-spaces-existence-tricky/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0CZH: focused-seeds/geometry/S9-spaces-existence-tricky/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In Situation 08B7. Let $\mathcal{K} \subset \mathcal{O}_X$ be a quasi-coherent sheaf of ideals. Let $X_e \subset X$ be the closed subspace cut out by $\mathcal{K}^e$. Let $\mathcal{I}_e = \mathcal{I}\mathcal{O}_{X_e}$. Let $(\mathcal{F}_n)$ be an object of $Coh_{\text{support proper over } A}(X, \mathcal{I})$. Assume*

1. *the functor $Coh_{\text{support proper over } A}(\mathcal{O}_{X_e}) \to Coh_{\text{support proper over } A}(X_e, \mathcal{I}_e)$ is an equivalence for all $e \geq 1$, and*
2. *there exists an object $\mathcal{H}$ of $Coh_{\text{support proper over } A}(\mathcal{O}_X)$ and a map $\alpha : (\mathcal{F}_n) \to \mathcal{H}^\wedge$ whose kernel and cokernel are annihilated by a power of $\mathcal{K}$.*

Then $(\mathcal{F}_n)$ is in the essential image of (08B8).

Proof. During this proof we will use without further mention that for a closed immersion $i : Z \to X$ the functor i_* gives an equivalence between the category of coherent modules on Z and coherent modules on X annihilated by the ideal sheaf of Z, see Cohomology of Spaces, Lemma 08AM. In particular we think of

$$Coh_{\text{support proper over } A}(\mathcal{O}_{X_e}) \subset Coh_{\text{support proper over } A}(\mathcal{O}_X)$$

as the full subcategory of consisting of modules annihilated by $\mathcal{K}^e$ and

$$Coh_{\text{support proper over } A}(X_e, \mathcal{I}_e) \subset Coh_{\text{support proper over } A}(X, \mathcal{I})$$

as the full subcategory of objects annihilated by $\mathcal{K}^e$. Moreover (1) tells us these two categories are equivalent under the completion functor (08B8).

Applying this equivalence we get a coherent $\mathcal{O}_X$-module $\mathcal{G}_e$ annihilated by $\mathcal{K}^e$ corresponding to the system $(\mathcal{F}_n/\mathcal{K}^e\mathcal{F}_n)$ of $Coh_{\text{support proper over } A}(X, \mathcal{I})$. The maps $\mathcal{F}_n/\mathcal{K}^{e+1}\mathcal{F}_n \to \mathcal{F}_n/\mathcal{K}^e\mathcal{F}_n$ correspond to canonical maps $\mathcal{G}_{e+1} \to \mathcal{G}_e$ which induce isomorphisms $\mathcal{G}_{e+1}/\mathcal{K}^e\mathcal{G}_{e+1} \to \mathcal{G}_e$. We obtain an object $(\mathcal{G}_e)$ of the category $Coh_{\text{support proper over } A}(X, \mathcal{K})$. The map α induces a system of maps

$$\mathcal{F}_n/\mathcal{K}^e\mathcal{F}_n \longrightarrow \mathcal{H}/(\mathcal{I}^n + \mathcal{K}^e)\mathcal{H}$$

whence maps $\mathcal{G}_e \to \mathcal{H}/\mathcal{K}^e\mathcal{H}$ (by the equivalence of categories again). Let $t \geq 1$ be an integer, which exists by assumption (2), such that $\mathcal{K}^t$ annihilates the kernel and cokernel of all the maps $\mathcal{F}_n \to \mathcal{H}/\mathcal{I}^n\mathcal{H}$. Then $\mathcal{K}^{2t}$ annihilates the kernel and cokernel of the maps $\mathcal{F}_n/\mathcal{K}^e\mathcal{F}_n \to \mathcal{H}/(\mathcal{I}^n + \mathcal{K}^e)\mathcal{H}$ (details omitted; see Cohomology

of Schemes, Remark 0888). Whereupon we conclude that $\mathcal{K}^{4t}$ annihilates the kernel and the cokernel of the maps

$$\mathcal{G}_e \longrightarrow \mathcal{H}/\mathcal{K}^e\mathcal{H},$$

(details omitted; see Cohomology of Schemes, Remark 0888). We apply Lemma 08BB to obtain a coherent $\mathcal{O}_X$-module $\mathcal{F}$, a map $a : \mathcal{F} \to \mathcal{H}$ and an isomorphism $\beta : (\mathcal{G}_e) \to (\mathcal{F}/\mathcal{K}^e\mathcal{F})$ in $Coh(X, \mathcal{K})$. Working backwards, for a given n the triple $(\mathcal{F}/\mathcal{I}^n\mathcal{F}, a \bmod \mathcal{I}^n, \beta \bmod \mathcal{I}^n)$ is a triple as in the lemma for the morphism $\alpha_n \bmod \mathcal{K}^e : (\mathcal{F}_n/\mathcal{K}^e\mathcal{F}_n) \to (\mathcal{H}/(\mathcal{I}^n + \mathcal{K}^e)\mathcal{H})$ of $Coh(X, \mathcal{K})$. Thus the uniqueness in Lemma 08BB gives a canonical isomorphism $\mathcal{F}/\mathcal{I}^n\mathcal{F} \to \mathcal{F}_n$ compatible with all the morphisms in sight.

To finish the proof of the lemma we still have to show that the support of $\mathcal{F}$ is proper over A. By construction the kernel of $a : \mathcal{F} \to \mathcal{H}$ is annihilated by a power of $\mathcal{K}$. Hence the support of this kernel is contained in the support of $\mathcal{G}_1$. Since $\mathcal{G}_1$ is an object of $Coh_{\text{support proper over } A}(\mathcal{O}_{X_1})$ we see this is proper over A. Combined with the fact that the support of $\mathcal{H}$ is proper over A we conclude that the support of $\mathcal{F}$ is proper over A by Derived Categories of Spaces, Lemma 0CZH. □